



FAR WESTERN UNIVERSITY
FACULTY OF AGRICULTURE SCIENCE
SCHOOL OF AGRICULTURE
Tikapur, Kailali

ECONOMIC ANALYSIS OF AGRICULTURAL RISK MANAGEMENT IN BANANA FARMING: A CASE STUDY OF CHITWAN DISTRICT

By

Hem Chand (40)

A case study report

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Far Western University, School of Agriculture

Asst Prof. Sijan Lamsal

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ABSTRACT

This case study analyze the economic efficiency of risk management strategies in commercial banana (*Musa spp.*) farming in the Chitwan District of Nepal specifically within the Ratnanagar and Kalika municipalities. The study evaluate the trade-offs between high-return commercial farming and the severe financial risks posed by abiotic stresses, primarily pre-monsoon windstorms. By utilizing a descriptive research design with secondary data from MoALD, the study estimates the cost of cultivation and performs a profitability analysis.

The findings reveal that while banana farming is a lucrative enterprise with a Benefit-Cost Ratio (BCR) of 1.47 under normal conditions and it remains highly vulnerable to climatic hazards. A scenario analysis demonstrates that a single windstorm event can result in a net loss of Rs 184,500 per Bigha for uninsured farmers, whereas insured farmers can secure a marginal profit even in disaster years. Despite the Government of Nepal providing an 80% subsidy on insurance premiums that making the marginal cost of safety (NRs 2,500) significantly lower than the potential benefit but it's adoption is constrained by institutional barriers. The study identifies the requirement for land qwnership certificates (*Lalpurja*) as a critical hurdle for tenant farmers (*thekka*). The report concludes by recommending policy reforms to recognize registered lease agreements (*Kabuliyatnama*) for insurance eligibility to ensure equitable risk transfer mechanisms for smallholder and tenant farmers.

Keywords: *Banana Farming, Agricultural Risk, Crop Insurance, Cost of Cultivation, Chitwan, Benefit-Cost Ratio.*

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1. INTRODUCTION

Agriculture plays a significant role in Nepal's economy contributing about 24.1 percent to the national GDP (MoALD, 2023). Over the past decade, the Terai region has seen a major shift from subsistence cereal farming toward high-value horticultural crops. Chitwan district located in the inner Terai. It has become one of the country's leading commercial banana-producing zones and is often referred to as the 'Banana Zone' under the Prime Minister Agriculture Modernization Project (PMAMP). Commercial banana cultivation is a 'high-input, high-reward' model than subsistence farming requiring substantial upfront investment in tissue-culture saplings, chemical fertilizers, and hired labor (Shrestha et al., 2018).

Commercial banana farming can earn more than crops like paddy and maize but it requires careful risk management to protect the investment from unpredictable weather. Even though the business is profitable, it still faces serious uncertainties. In Chitwan, the biggest problem is abiotic stress especially strong windstorms that hit during the pre-monsoon season. The banana plant's biological structure heavy fruit bunches supported by a shallow root system makes it inherently susceptible to lodging (toppling) during these storms (Ghimire et al., 2016). The economic consequences are extremely serious in a 2005 windstorm destroyed 400 hectares in Chitwan. It causing losses exceeding Rs 29 million and pushing farmers into long-term debt traps (World Bank, 2009).

For mitigate these risks, the Government of Nepal introduced The Crop and Livestock Insurance Directive (2013). It offering an 80% premium subsidy but even with the subsidy many farmers still don't join. The main reason is the requirement of Land Ownership Certificates (Lalpurja) for insurance eligibility. This policy systematically excludes tenant farmers (thekka) who make up a large share of banana growers. This study analyzes the economic trade-offs of these risks and the efficiency of current safeguarding mechanisms.



2. OBJECTIVES OF STUDY

1. To identify the major agricultural risks faced by commercial banana farmers in Chitwan District with specific focus on production risks such as windstorms, biotic stresses and market fluctuations.
2. To estimate the cost of cultivation, gross margin, and Benefit-Cost Ratio (BCR) per Bigha under both normal (risk-free) and hazard affected farming conditions.
3. To find the economic efficiency of safeguarding measures specifically physical propping and the Government of Nepal's crop insurance scheme by comparing their costs and financial benefits during risk events.
4. To identify the institutional barriers particularly land tenure issues that hindering the widespread adoption of insurance among tenant farmers.
5. To find the cost of cultivation and gross margin per bigha under risk-free between risk-prone scenario and evaluate the economic efficiency of safeguarding measures and comparing the cost-benefit of physical propping versus crop insurance.

3. METHODOLOGY

3.1 Selection of Study Area

Chitwan District was selected for this study due to its status as the largest banana-producing district in Bagmati Province within Ratnanagar Municipality and Kalika Municipality were chosen as the specific study sites. These areas have the highest density of commercial banana plantations and are geographically exposed to wind corridors so that it making them ideal for risk analysis.

3.2 Data Collection Sources

Data for this study were collected from various secondary sources like website and documents -

- *Statistical Information on Nepalese Agriculture (2022/23)*, MoALD.
- *Journal of the Nepal Agricultural Research Council*.
- Directives from the *Nepal Insurance Authority* (Beema Samiti)
- *Chitwan Banana Producers Association*



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3.3 Analytical Framework

The collected data were analysed using following economic indicators:

A. Gross Margin Analysis:

$$\text{Gross Margin (GM)} = \text{Total Revenue (TR)} - \text{Total Variable Cost (TVC)}$$

B. Benefit-Cost Ratio (BCR):

$$\text{BCR} = \text{Total Profit} / \text{Total Cost}$$

C. Risk Assessment: Risks are categorized based on Frequency (how often they occur?) and Severity (economic loss per occurrence).

4. AREA OF STUDY

Chitwan located in the central southern part of Nepal (Inner Terai). The climate is subtropical with hot, humid summers and cool dry winters. In this region, the average annual rainfall is roughly 2,000 mm which is mostly concentrated in the monsoon season (June–September). The soil texture in Ratnanagar and Kalika is predominately sandy loam to loam also well-drained and ideal for banana cultivation. However, the open topography leaves the crop vulnerable to high winds during the pre-monsoon squalls.. The farming system has shifted from integrated subsistence (crop-livestock) to specialized monoculture. Most of these commercial farming is done on leased land (thekka) where farmers rent land for 5 to 10 years paying an annual rent of NRs 40,000 - 60,000 per Bigha.

5. FINDINGS AND DISCUSSION

Commercial banana farmers in Chitwan are mostly male about 85% and the average age is 45 years. These commercial farmers have more educated than subsistence farmers which helps them understand complicated inputs like tissue-cultured plants and insurance policies. Study data found that on average each commercial banana farmer owns about 2 to 5 Bigha of land. Data from the study provided three main types of risks that farmers face in banana farming.



Table 1: Risks based on their potential to cause financial ruin

Rank	Risk Type	Specific Threat	Impact Description
1	Production (Abiotic)	Windstorms	Immediate lodging of fruiting trees; 100% loss of yield.
2	Production (Biotic)	Panama Disease Wilt	Long-term soil infection; forces variety change or land abandonment.
3	Market Risk	Price Fluctuations	Influx of cheaper Indian bananas lowers local farm-gate price.

Source: Field Survey & KII, 2025.

To understand the impact of risk, we first establish the baseline Cost of Cultivation for **1 Bigha (0.67 ha)** of Banana (*William Hybrid* variety).

Here a data of estimated cost of cultivation at normal season:

(Appendix A)

5.1 Profitability Analysis (Risk-Free Scenario)

Under normal weather conditions with no major events:

Average Yield	Avg. Price	Total Revenue	Total Cost	Net profit	Benefit Cost Ratio
1,100 bunches per Bigha (<i>allowing for 8-10% less</i>)	Rs. 350 per bunch	$1100 \times 350 =$ Rs. 385000	Rs. 261,500	Rs. 385,000 - Rs. 261,500 = Rs. 123,500	1.47

This indicates that under safe conditions, banana farming is a lucrative business returning 1.47 for every rupee invested.

5.2 Economic Analysis of Safeguarding Measures

5.2.1 Physical Safeguarding (Propping) : Propping involves supporting banana bunches with bamboo sticks or plastic ropes. The cost of propping is around Rs 32,000 per Bigha which accounts for approximately 13% of the total production cost. It is effective in protecting the plants against mild to moderate winds of up to 30-40 km/h but it cannot withstand cyclonic



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storms. Propping being a relatively expensive but it's necessary. Even a normal summer breeze can cause the heavy William Hybrid banana bunches if they don't have proper support.

5.1.2 Financial Safeguarding (Crop Insurance) : The Government of Nepal offers an insurance scheme that covers risks which can't control such as wind, floods, and diseases. The insurance amount is based on the cost of cultivation which is around Rs 250,000 per Bigha. The premium is 5% of that amount which is around Rs 12,500. The government pays 80 percent of the premium which is Rs 10,000. So the farmer only needs to pay Rs 2,500 to join the scheme.

5.2.3 Scenario Analysis (The 2025 Windstorm Simulation) :

Let us assume a severe windstorm destroys 80% of the crop.

Table 3: Comparison of Farmers based on severe indicators

Indicator	Farmer A (No Insurance)	Farmer B (Insured)
Total Investment	Rs 261,500	Rs 264,000 (includes premium)
Yield Revenue (20% survival)	Rs 77,000	Rs 77,000
Insurance Claim Payout	Rs 0	Rs 200,000 (80% of Sum Insured)
Total Income	Rs 77,000	Rs 277,000
Net Profit/Loss	(Rs 184,500) LOSS	Rs 13,000 PROFIT

Discussion: without insurance, a single event wipes out the farmer's capital. It leading to a debt trap. The farmer recovers their entire input cost and makes a marginal profit with insurance. It allowing them to farm again the next season. The marginal cost of insurance (Rs 2,500) provides a marginal benefit of Rs 200,000 in the event of a disaster.

5.3 Constraints in Adoption

Despite the clear economic benefit, adoption in Chitwan is hindered by Institutional Risk:

Land Tenure (*theikka*): Insurance companies require the Lalpurja (Land Ownership Certificate) to process policies. Those tenant farmers (who lease land verbally or via unregistered contracts) struggle to provide this.

Valuation Method: The current insurance pays based on input cost not market value. Commercial farmers argue that they lose the potential profit not just the cost.



6. CONCLUSION

Banana farming in Chitwan is a high-stakes economic activity. While the **Benefit-Cost Ratio of 1.47** indicates high profitability. The variance in returns due to wind risks is extreme. The study concludes that windstorms are the primary threat to the long-term economic sustainability of banana farms in the district. Propping helps reduce losses and is a necessary production cost but not sufficient for disaster management. Crop insurance remains the most efficient risk-transfer tool. It offering a good return on investment during years of heavy damage. The problem is that current land ownership requirements do not reflect the realities of tenant farming, which prevents the most vulnerable farmers from accessing this protection.

7. RECOMMENDATIONS

1. The Insurance Board must mandate that a registered Lease Agreement (*Kabuliyatnama*) is sufficient for insurance purposes that help to removing the mandatory requirement for *Lalpurja* from the land owner.
2. The government should pilot revenue insurance products that cover not just the cost of inputs but the expected market value of the crop. It protecting the farmer's livelihood not just their capital.
3. Insurance should be compulsory bundled with bank loans for commercial agriculture to ensure 100% coverage of credit-linked farmers.



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APPENDIX

Appendix A.

Table 2: Estimated Cost of Cultivation per Bigha (Normal Season)

S.N	Cost Head	Unit	Rate (NRs)	Total Cost (NRs)	% Share
A	Variable Costs				
1	Saplings (Tissue Culture)	1,200 plants	25	30,000	12.2%
2	Fertilizers (DAP, Urea, Potash, Boron)	Lumpsum	-	55,000	22.4%
3	Organic Manure (FYM/Chicken Manure)	4 Trolley	6,000	24,000	9.8%
4	Labor (Land prep, weeding, harvest)	60 Man-days	800	48,000	19.6%
5	Plant Protection (Fungicides/Pesticides)	Lumpsum	-	15,000	6.1%
6	Safeguarding: Propping (Bamboo/Rope)	800 pcs	40	32,000	13.1%
7	Safeguarding: Insurance Premium (20% share)	Lumpsum	-	2,500	1.0%
8	Irrigation & Fuel	Lumpsum	-	10,000	4.1%
	Total Variable Cost (TVC)			216,500	
B	Fixed Costs				
1	Land Rent (<i>thekka</i>)	1 Year	45,000	45,000	18.4%
	Total Cost of Production (TCP) = A+B			261,500	100%

Note: The Insurance Premium is calculated based on the Cost of Production valuation. GoN pays 80%, Farmer pays 20%.

(Source: Field Survey, 2025)



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